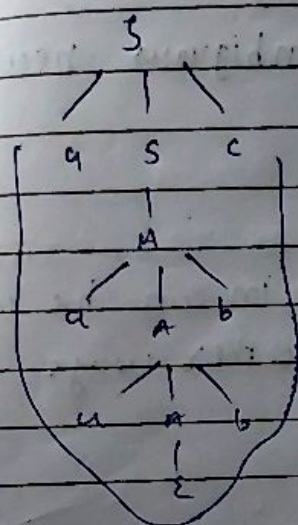


9) Find context-free grammars for the language:
 $L = \{a^i b^j c^k \mid i = j + 1\}$

→ $S \rightarrow aSc \mid A$
 $A \rightarrow aAb \mid \epsilon$



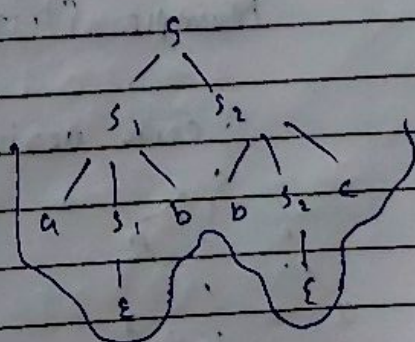
$\Rightarrow a a a \epsilon b b c$
 $\Rightarrow a^3 b^2 c^1$

10) Find context-free grammars that generates the following languages:-

→ i) $L_1 = \{a^i b^j c^k \mid i = j + 1\}$
 can be

ii) $L_2 = \{a^i b^j c^k \mid j = i \text{ or } j = k\}$

$S \rightarrow S_1 \mid S_2$
 $S_1 \rightarrow a S_1 b \mid \epsilon$
 $S_2 \rightarrow b S_2 c \mid \epsilon$



$a b b c$
 $a^1 b^2 c^1$